Annex — Virginia Law Officers (VaLORS) pre-score

A priced amendment held in reserve · Policy Director · MOO-100 · 12 October 2026 · internal, pre-decisional

Status: reserve document. Not part of the bill, not part of the fiscal note, not for circulation. The board decided on 12 October 2026 to exclude the State Police Officers', Virginia Law Officers' and Judicial Retirement Systems from the pilot, expressly in the bill text. On 12 October 2026 the board further decided to price a VaLORS amendment now, before it is offered on the floor, rather than after. This annex is that price. It is produced so that the answer to "what would it cost to add law officers?" is a number we already hold, not a number the other side supplies.

Built on the same method as V2-fiscal-model, from the same source of record, with the same three dials at the same settings. Only the plan changes. Figures are computed by the same code, so this annex and the model cannot drift apart.

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0. The answer in four lines

	Low	Mid	High
VaLORS participants added over the four-year pilot	0.6	**2.8**	9.5
Member payments into the trust fund	$11,315	**$51,726**	$175,113
Trust-fund subsidy at k = 2.0	$11,315	**$51,726**	$175,113
Increase over the bill as scoped — participants	+3.5%	**+6.7%**	+10.9%
Increase over the bill as scoped — member payments	+4.9%	**+9.3%**	+15.1%

The amendment is small, and the reason is the plan's own eligibility rules, not our drafting. VaLORS is a young, high-turnover workforce — average age 39.6, average total service 8.3 years — and its unreduced-retirement rules are age 60 with five years or age 50 with 25 years. Only 583 of 7,459 members (7.8%) sit in either pathway. The comparable figure for the State Employees plan is 8,650 of 28,994 (29.8%) — nearly four times the rate.

So the sympathetic amendment is also an affordable one, and that is the single most useful sentence in this annex. At mid take-up, adding 7,459 law officers to the pilot adds about 3 participants and about $51,726 of member payments over four years. We should be able to say that in a committee room without reaching for a calculator.

What we cannot say is what it costs the trust fund, for the same reason we cannot say that for the bill as scoped: the actuarial-equivalent-to-approximate-normal-cost ratio, k, is unknown and only VRS can supply it. Everything below is expressed in member payments and in k.

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1. Why VaLORS is structurally different from the plan we scoped

1.1 There is no VaLORS Hybrid — the whole plan is in scope

This is the finding that most changes how a VaLORS amendment should be read. Hazardous-duty positions are not eligible for the Hybrid Retirement Plan. A VaLORS member whose membership date is on or after 1 July 2010 is in **Plan 2**, even if the date is after 31 December 2013. So all 7,459 active VaLORS members are Plan 1 or Plan 2, and a bill naming Plan 1 and Plan 2 reaches every one of them.

In the State Employees plan the Hybrid tier absorbs 52,280 of 81,274 members — 64% — and the Plan 1/Plan 2 denominator is only 28,994. **A reader who assumes the same 36% in-scope fraction applies to VaLORS will understate the VaLORS denominator by about 2.8×.** It does not change the answer in section 0, because the binding constraint is eligibility and not tier, but it is the kind of error that gets made out loud.

Table 1 — plan-tier split, derived. Hazardous duty has no Hybrid tier, so one cut at 15.0 years of total service separates the tiers and it falls exactly on a published band boundary. No band has to be split.

Tier	Members	Average salary	Basis
VaLORS Plan 1	1,384	$62,721	Hired before 1 July 2010 → total service ≥ 15.0 yr at 30 June 2025
VaLORS Plan 2	6,075	$56,938	Everyone else. No Hybrid tier exists for hazardous duty
Total — the whole plan is in scope	**7,459**	$58,011	VRS valuation, p. 49

1.2 The eligibility rules are different, and they are what makes the amendment small

VaLORS Plan 1 and Plan 2 share one rule set: an unreduced allowance at **age 60 with at least five years of service credit, or at age 50 with at least 25 years of service credit**. That is materially more generous than the State Employees rules, and it cuts in an unexpected direction for this bill.

- **The age-60 route cannot be purchased into.** Once a member has five years, buying service does nothing for eligibility; before five years, the hazardous-duty rule requires five years of VaLORS, SPORS or enhanced hazardous-duty service, which purchased military service does not supply. Section 1.3 treats this as the governing reading.
- **The age-50-with-25-years route can be purchased into**, and it is the only Bridge route in VaLORS. It reaches members already aged 50 or over who hold 21 to just under 25 years of service.
- **Almost nobody is in that box.** A workforce averaging 8.3 years of service has very few members with 21 to 25 years and age 50 or more. The published table gives 150.

Table 2 — the two pathway pools, derived cell by cell from the published VaLORS age × total-service table

Pool	Count	Average salary	Composition
Retirement Incentive — already eligible for an unreduced allowance	433	$63,701	Plan 1 319 (73.7%) · Plan 2 114
Retirement Bridge — reaches the threshold with ≤ 48 months purchased	150	$63,996	Plan 1 150 · Plan 2 0
Combined pathway-eligible	**583**	$63,776	**7.8% of 7,459** — against 29.8% for State Employees

Bridge members need an average of **24.0 months**, the same figure the State Employees model produces and for the same reason: the whole Bridge pool sits inside one published service band, so a uniform assumption inside that band yields a uniform months-needed distribution by construction. The average is robust. The shape is an artefact and we do not defend it.

1.3 The VaLORS Bridge is Plan 1 only, and the strict screen governs

Expected VaLORS Plan 2 Bridge participation is zero for the whole pilot, by exactly the argument the board adopted for the State Employees plan on 12 October 2026. A VaLORS Plan 2 member holds at most 15 years of service at 30 June 2025 and at most about 20 by 2031, so the 25-year route is out of reach even with 48 months purchased. Plan 2 should stay legally eligible in the drafting and be scored at zero in the note.

There is a second, wider reading of the Bridge that we are **not** adopting. Members aged 60 or over with one to just under five years of service could in principle buy their way to the five-year threshold and retire unreduced at once. That reading adds 75 members and would take the Bridge pool to 225.

Table 3 — the two VaLORS Bridge screens

Screen	Plan 1	Plan 2	Total	Status
Strict — service-route only (age 50 + 25 years)	150	0	**150**	**Governs this annex**
Wider — also allows buying to the five-year vesting threshold at age 60+	150	75	225	Not adopted. Open question for VRS

We take the strict screen for a stated reason: VRS requires five years of service **as a VaLORS, SPORS or enhanced hazardous-duty member** to retire with hazardous-duty benefits, and purchased military service does not count toward that requirement. A member who bought their way to five years of total creditable service would still not have five years of hazardous-duty service, and so would not reach the age-60 hazardous-duty rule. We are confident of the rule and not certain of its interaction with a purchase, so we take the smaller pool and put the question to VRS. **If VRS says the wider reading is right, the Bridge pool rises by 50% and total VaLORS participants rise by about 13%.** That is the sensitivity, and it does not change the order of magnitude.

This is the same discipline the board applied to the State Employees Bridge screen: take the smaller defensible pool, state the larger one, and name the question that resolves it.

1.4 One piece of good news that belongs in the fiscal note

Purchased military service does not create hazardous-duty supplement eligibility. VRS is explicit: other types of prior service, including active-duty military service, do not count toward the 20 years of hazardous-duty service the supplement requires. So a VaLORS amendment cannot manufacture a supplement obligation, and the supplement — a dollar amount paid monthly from retirement to age 65, available only to pre-1 July 2001 members on the 1.7% multiplier — is outside this amendment's cost entirely. Say that before VRS has to ask.

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2. Price — the 1.9× headline is wrong, and the real multiple is about 1.4×

The approximate-normal-cost rate for hazardous-duty positions is 23.78% of salary for Plan 1 and 19.15% for Plan 2, against 12.50% and 10.74% for non-hazardous state positions, all effective 1 July 2025. That is where the “roughly 1.9× the price per year purchased” line in V2-fiscal-model comes from, and as a statement about **rates** it is right.

As a statement about **cost per participant** it is wrong, because VaLORS salaries are lower. Average VaLORS compensation is $58,011 against $77,215 for State Employees — 25% lower — and the member payment is charged on salary. The two effects partly cancel.

Table 4 — member payment per participant, VaLORS against the plan we scoped

Pathway	Years purchased at ANC	Rate	Average salary	Member pays	State Employees	Multiple
Retirement Incentive (12-month cap, G3/G18)	1.00	22.56% blended	$63,701	**$14,372**	$10,514	1.37×
Retirement Bridge (months needed only, G2/G18)	2.00	23.78%	$63,996	**$30,436**	$20,806	1.46×

Correct the 1.9× line wherever it appears as a cost statement. Per year of service purchased the hazardous-duty rate is 1.90× the Plan 1 non-hazardous rate. Per participant the cost runs about 1.37× on the Incentive and 1.46× on the Bridge. Anyone who argues “nearly twice as expensive per veteran” is overstating it by about a third, and that is an argument we will be on the wrong side of if we make it ourselves first.

Table 5 — trust-fund subsidy at four values of k, mid case (2.8 VaLORS participants)

k	Actuarial-equivalent value of service granted	Member pays	Trust-fund subsidy
1.5	$77,588	$51,726	$25,863
2.0	$103,451	$51,726	$51,726
2.5	$129,314	$51,726	$77,588
3.0	$155,177	$51,726	$103,451

k is probably higher for VaLORS than for the plan we scoped, and in the one direction we cannot measure. Hazardous-duty members reach an unreduced allowance up to five years earlier, and the actuarial-equivalent cost of a purchase rises with how early the resulting allowance starts. VaLORS total normal cost is 14.30% of payroll against 8.73% for State Employees, which is the same fact seen from the employer side. We do not have a VaLORS k and we are not going to invent one. **If a VaLORS amendment is offered, k for VaLORS becomes a second actuarial question, not a scaling of the first.**

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3. Reach — the same three dials, at the same settings

Veteran share, share holding qualifying purchasable service, and take-up, exactly as in V2-fiscal-model. The dials are held at the same values deliberately, so the VaLORS column is comparable with the State Employees column rather than being argued separately.

Table 6 — take-up only, veteran share 8% and qualifying share 50%

Take-up	Incentive	Bridge	Total	Member payments	Subsidy at k = 2.0
5%	0.9	0.3	1.2	$21,552	$21,552
12%	2.1	0.7	2.8	$51,726	$51,726
25%	4.3	1.5	5.8	$107,762	$107,762

Table 7 — all 27 cells. Participants added over the four-year pilot.

Veteran share	Qualifying share	Take-up 5%	Take-up 12%	Take-up 25%
6%	35%	0.6	1.5	3.1
	50%	0.9	2.1	4.4
	65%	1.1	2.7	5.7
8%	35%	0.8	2.0	4.1
	50%	1.2	2.8	5.8
	65%	1.5	3.6	7.6
10%	35%	1.0	2.4	5.1
	50%	1.5	3.5	7.3
	65%	1.9	4.5	9.5

Honest full range: 0.6 to 9.5 VaLORS participants over four years, $11,315 to $175,113 in member payments. Two of the three dials are data requests and not judgments. The VaLORS-specific versions of both are now added to the G24 request.

3.1 The one dial that should probably be set higher for VaLORS

Veteran share is almost certainly higher among law officers than across the classified state workforce, and we hold no Virginia figure for it. The 6% we use as a floor comes from DHRM's January 2025 presentation and covers the classified workforce as a whole. Corrections and law-enforcement roles draw disproportionately from people with military service. We will not put a number on that from a secondary source, so the table below is labelled as what it is: arithmetic on a judgment, offered so the sensitivity is visible rather than hidden.

Table 8 — veteran-share sensitivity. Qualifying share 50%, take-up 12%. The multipliers are judgment with no source. Do not quote them as findings.

Veteran share	Basis	VaLORS participants	Member payments
8%	Base — same as the State Employees model	2.8	$51,726
12%	1.5× the base. Judgment, no source	4.2	$77,588
16%	2.0× the base. Judgment, no source	5.6	$103,451
20%	2.5× the base. Judgment, no source	7.0	$129,314

Even at 2.5× the veteran share, a VaLORS amendment adds about 7 participants at mid take-up. That is the robust point. The amendment is small across every setting of the dial we are least sure about, because the binding constraint is eligibility and not veteran status.

3.2 Where this annex is conservative, stated plainly

Both screens ask who is eligible, or reachable, **at 30 June 2025**. The pilot would run four years. Members age and accrue service throughout. Rolling the published table forward on both axes gives

the scale of what the static screens leave out.

Table 9 — roll-forward. Nothing leaves VaLORS in this view, so these are upper bounds on the dynamic effect, not projections.

View	Incentive pool	Bridge pool	Combined	Against the static screen
Static — 30 June 2025 (governs this annex)	433	150	583	—
+2 years — mid-pilot	533	168	701	+20%
+4 years — end of pilot	634	185	819	+41%

This conservatism applies to the State Employees model too, where it had not been stated. Our published pools of 6,254 and 2,396 are also a 30 June 2025 snapshot. The direction of the error is known — we understate — and the size is of this order. It belongs in the fiscal note as a stated limitation rather than as something a reviewer finds. It is now carried in three places: **§6.1 of V2-fiscal-model**, added as amendment 1 of 12 October 2026; **item 5 of the data request**, which states it to VRS and asks for the counts projected to the pilot years; and open item 7 in §7 below.

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4. The workforce ledger (G11 Ledger 2) — and the amortization trap is worse here

Separate ledger, agency appropriations, never netted against the pension cost. Decision 6 of 12 October 2026 holds without exception and a VaLORS amendment does not reopen it.

Table 10 — the VaLORS employer rate, and which parts are a real saving when payroll falls

Component	VaLORS	State Employees	A saving when payroll falls?
Employer normal cost	9.30%	4.34%	Yes
Administrative expense	0.26%	0.24%	Yes
Amortization of unfunded liability	**13.04%**	6.49%	**No** — see below
Employer DB contribution rate	22.60%	11.07%	
FICA	7.65%	7.65%	Yes
Honest scalable multiplier	**1.1721**	1.1223	

For VaLORS, 13.04 of the 22.60 points — 58% of the employer rate — is amortization of a fixed $717.570 million unfunded liability, and none of it is a saving. Cutting VaLORS payroll does not reduce that liability by a cent; it reduces the dollars collected against it and the rate must rise later. The trap is the same one the State Employees ledger carries, and it is twice as large as a share of the rate. Anyone presenting 22.60% as avoided employer cost would be booking a timing cost as a saving.

Table 11 — VaLORS agency payroll saving, annual run-rate at mid take-up (2.8 retirements, blended salary $63,776)

Positions not refilled	Replacement at 80%	at 85%	at 90%
1%	$43,481	$33,133	$22,786
3%	$46,826	$36,687	$26,549
5%	$50,170	$40,241	$30,311

Table 12 — where the mid-cell saving comes from (85% replacement, 3% not refilled)

Source	Annual	Share
Salary compression on replacement	$30,416	82.9%
Positions not refilled	$6,271	17.1%
Total	**$36,687**	
Memo — amortization dollars no longer collected. A timing cost, not a saving.	$4,082	

Not modelled, and the same gaps as the main ledger: state health insurance per head, leave payout at separation, vacancy and training cost during transition. For VaLORS there is a further one that matters more than it does elsewhere — **a corrections or law-enforcement post that is not refilled may not be a saving at all** if it is backfilled with overtime. We hold no figure for that and it would cut against us. It belongs in the note.

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5. State Police (SPORS) — the amendment that usually travels with it

Scored here for the same reason, at the same settings. SPORS shares the VaLORS eligibility rules — age 60 with five years or age 50 with 25 years — and the same hazardous-duty approximate-normal-cost rates. It is a much smaller plan with much higher salaries, and those two facts almost cancel in the dollars.

Table 13 — SPORS pre-score

	SPORS	VaLORS
Active members	1,902	7,459
Average compensation	$96,071	$58,011
Incentive pool	268	433
Bridge pool (strict screen)	60	150
Pathway-eligible share of the plan	17.2%	7.8%
Member pays — Incentive, per participant	$29,713	$14,372
Member pays — Bridge, per participant	$58,544	$30,436
Participants, mid take-up	1.6	2.8
Member payments, mid take-up	**$55,084**	**$51,726**
Participants, full range	0.3 – 5.3	0.6 – 9.5
Employer amortization share of the DB rate	17.96% of 31.67% — 57%	13.04% of 22.60% — 58%

SPORS is a quarter of the size of VaLORS and costs about the same, because the salaries are double. Pathway-eligibility runs at 17.2% against 7.8% for VaLORS — state police serve longer careers — so the eligible fraction is higher even though the plan is far smaller. One more fact worth holding: SPORS members must retire within 60 days of age 70, and no active SPORS member is aged 69 or over in the published table. The Incentive pathway cannot reach anyone the mandatory-retirement rule would have reached anyway.

Table 14 — what each amendment adds at mid take-up

Scope	Participants	Member payments	Against the bill as scoped
Bill as scoped — State Employees Plan 1/Plan 2 only	41.5	$554,922	—
+ VaLORS	44.3	$606,648	+6.7% participants, +9.3% payments
+ VaLORS and SPORS	45.9	$661,732	+10.5% participants, +19.2% payments

5.1 Judicial (JRS) is deliberately not scored

We should not put a number on JRS and this annex does not. The Judicial Retirement System applies weighted service credit at rates that depend on the date of appointment, which means “years of service credit” in the JRS eligibility rules is not the same quantity as the total service published in the valuation table. Applying the VaLORS method to the JRS table would produce a confident-looking figure built on the wrong variable.

What we can say: JRS has 473 active members, the oldest average age in the system at 57.3, and an approximate-normal-cost rate of 31.97% for Plan 1 and 29.67% for Plan 2 — about 2.6× the Plan 1

non-hazardous rate — on an average compensation of $208,551. **Per participant, a judicial purchase is by far the most expensive in the system.** If a JRS amendment is actually offered, the weighted-service rules have to be verified first. That is a day of work and it is listed as an open item rather than guessed at here.

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6. How to use this annex

1. **It stays in reserve.** It is not attached to the bill, the fiscal note, the one-pager or the advocacy brief. The bill as drafted expressly excludes SPORS, VaLORS and JRS, and that remains the position.
2. **If a VaLORS amendment is offered, the answer is a number and not a flinch.** At mid take-up it adds about 3 participants and about $51,726 in member payments over four years — roughly 9% on top of the bill as scoped.
3. **Lead with why we excluded them, not with the cost.** The reason is sequencing: prove the mechanism on the one plan we can model from published data, then extend. The cost is modest and saying so first makes the sequencing argument sound like a pretext.
4. **Do not accept a floor amendment that adds a plan without adding k for that plan.** A VaLORS amendment needs its own actuarial ratio. Accepting one without it converts a bounded, priced pilot into an unpriced one, which is the specific failure mode G13 exists to prevent.
5. **If the board wants VaLORS in the bill from the start, say so before 30 November 2026.** It is a different bill with a different fiscal note, and the data request would need a VaLORS section that it does not currently have. Adding it at the drafting request is cheap; adding it in February is not.

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7. Open items this annex creates

#	Item	Owner	Why it matters
1	Does purchased military service let a VaLORS member aged 60+ reach the hazardous-duty five-year threshold, or does that threshold require five years of hazardous-duty-covered service?	VRS	The wider Bridge screen. Worth +75 members, +13% of VaLORS participants
2	k for VaLORS Plan 1 and Plan 2, by age band	VRS actuary (GRS)	Probably above the State Employees k. Cannot be scaled from it
3	Veterans among active VaLORS and SPORS members	VRS / DHRM	The dial we are least sure about and most likely understating
4	Members with qualifying purchasable military service, VaLORS and SPORS	VRS	Same weakest-link status as in the main model
5	Whether an unfilled corrections or law-enforcement post is backfilled with overtime	DHRM / Department of Corrections	Could reduce the Ledger 2 saving to near zero. Cuts against us
6	JRS weighted-service-credit rules by appointment date	Policy Director	Required before any JRS figure can be produced
7	Roll-forward limitation stated in the main fiscal note	Policy Director	Our State Employees pools are also a 30 June 2025 snapshot and understate by roughly this order

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8. Sources

All retrieved 12 October 2026. Primary Commonwealth sources only, per G25.

Source	Used for
VRS, *Annual Actuarial Valuation, June 30, 2025* — p. 49 Active Participant Detail; p. 54 Active Participants by Age and Total Service, Virginia Law Officers; p. 52 same table, State Police; p. 53 same table, Judicial; p. 16 Results Summary, Virginia Law Officers; p. 18 Employer Contribution Rate Detail	Every count, salary, employer rate and the $717.570M VaLORS UAAL
VRS, *Purchase of Prior Service Cost Rates for VRS Plan 1* and *for VRS Plan 2*, effective 1 July 2025	23.78% / 19.15% hazardous; 31.97% / 29.67% judicial; 12.50% / 10.74% non-hazardous
VRS, *Plan 1: Virginia Law Officers' Retirement System* overview, 04-2026	Unreduced eligibility age 60/5 and age 50/25; 1.7% and 2.0% multipliers; 36-month AFC; hazardous-duty supplement rules; prior service counts toward eligibility
VRS, *Plan 2: Virginia Law Officers' Retirement System* overview, 04-2026	Identical unreduced eligibility; 2.0% multiplier; 60-month AFC
VRS, *Hazardous Duty Benefits FAQ for Members*, April 2026	Eligibility rules confirmed for SPORS and VaLORS together; the five-years-of-hazardous-duty requirement; purchased military service excluded from the 20-year supplement test; SPORS mandatory retirement at 70
VRS, Hybrid Retirement Plan eligibility — hazardous-duty exclusion	No VaLORS or SPORS Hybrid tier; membership on or after 1 July 2010 is Plan 2 regardless of date
DHRM, presentation to Senate Finance and Appropriations, 21 January 2025, slide 8	Veterans 6% of the classified state workforce, as of 30 June 2024. Used as a floor

Derived, not published: the plan-tier split; both pathway pools; all average salaries by cohort; the months-needed distribution; the roll-forward. **Assumptions, labelled per G23:** veteran share; qualifying-service share; take-up; uniform distribution within published age and service bands; salary as a proxy for average final compensation; k as a ratio rather than a value. **Judgment with no source, and marked as such in Table 8:** the veteran-share multipliers for law-enforcement cohorts.

Nothing in this annex is a VRS actuarial result, and it contains no claim that any part of the proposal is actuarially neutral, minimal in impact or self-funding.